

Dr Praveen Kumar

(PhD. Biochemistry from CCSHAU)

5 Years work experience in teaching, research and diagnostics corporates.

Worked at Vivekanand degree college as an Assistant Professor (Jan-2020 to April-2021)

Worked at Patanjali Ayurveda as a R&D executive (May-2021 to July-2021)

Currently working at Thyrocare Technologies Ltd (From August-2021)

Designation : Lab Officer Quality control and Assurance; handling the analysers such as AU5800, Immulite, Unicell dxi800, Centaure, Beckman Access, Autobio, Cobas8000, Cobaspro, Sysmex XN, urine analyser, HPLC, GC and ICPMS.

Contact at 9991054596, praveenhau@gmail.com

Career Objective:

To pursue a career in an eminent progressive organization where I can utilize my knowledge and skills to the fullest and which can give me high opportunities for success and career development thereby contributing towards the growth of the organization.

Educational Qualification:

Degree	Board/University	Maximum marks	Marks Obtained	Percentage	Division	Subjects
10th	Haryana Board	600	415	69.17	First	Hindi, English, Mathematics, Science, Social science, Physical education
12th	Haryana Board	500	300	60.00	First	Hindi, English, Biology, Chemistry, Physics
B.Sc.	Bangalore University	3200	2315	72.34	First	Biochemistry, Biotechnology, Genetics
M.Sc.	Bangalore University	2000	1211	60.55	First	Biochemistry
Ph.D.	C.C.S.H.A.U.	CGPA -10	7.24	72.40	First	Biochemistry

Publications

Books

1. Kumar, P. (2012). Mobile phone causes cancer, Lambert academic publishing, ISBN 978-3-659-11970-5.

Research Papers

➤ Main Authored

1. Kumar, P. (2022). Measurement of Ascorbate Peroxidase Activity in Sorghum. Bio-protocol 12(20): e4531. DOI: 10.21769/BioProtoc.4531.

2. Kumar, P. (2022). Nanomaterial's synthesis, types and their use in Bioremediation and Agriculture. *Natural Resources for Human Health*, 2(3), pp.349-365.
<https://doi.org/10.53365/nrfhh/144289>
3. Kumar P (2021) Stress amelioration response of glycine betaine and *Arbuscular mycorrhizal fungi* in sorghum under Cr toxicity. *PLOS ONE* 16(7): e0253878.
<https://doi.org/10.1371/journal.pone.0253878>
4. Kumar, P. (2021). Soil applied glycine betaine with Arbuscular mycorrhizal fungi reduces chromium uptake and ameliorates chromium toxicity by suppressing the oxidative stress in three genetically different Sorghum (*Sorghum bicolor* L.) cultivars. *BMC Plant Biol* 21, 336,
<https://doi.org/10.1186/s12870-021-03113-3>.
5. Kumar, P. (2020). Evaluation of Toxic Effects of Hexavalent Chromium on the Yield and Quality of Sorghum. *Enliven Journal of Dietetics Research and Nutrition*, 7(2), pp.1-11.
6. Kumar, P., Tokas, J. and Singal, H.R. (2019). Amelioration of Chromium VI Toxicity in Sorghum (*Sorghum bicolor* L.) using Glycine Betaine. *Scientific reports*, 9(1), pp.1-15.
7. Kumar, P., Tokas, J. and Singal, H.R. (2018). Climate change consequences and its impact on agriculture and food security (A Review). *International Journal of Chemical Studies*, 6(6): 124-133.
8. Kumar, P. (2014). Role of Gluten protein in the food products of living beings and its effect on their body both physicochemical and metabolic reactions. *International Research Journal of Commerce Arts and Science*, 5(3): 65-83.

➤ **Co-Authored**

9. Kumar, N., Kumar, P., Tokas, J. and Singal, H. R. (2018). Post-harvest Morpho-physiological Changes in Tomato after Exogenous Application of Salicylic Acid. *Annals of Biology*, 34 (1): 92-94.
10. Kumar, A., Lal, M., Kumar, P., Kumar, R. and Kumar, J. (2018). Phenological and Growth Responses of Indian Mustard (*Brassica juncea* L.) Genotypes to Different Sowing Dates. *Int. J. Curr. Microbiol. App. Sci* 7(2): 1435-1440.
11. Lal, M., Sangwan, A., Kumar, P., Dinesh and Sheokand, S. (2017). Effects of Waterlogging, Salinity and their Combination on Seed Germination and Vigour Index in Pigeonpea [*Cajanus cajan* (L.) Mill sp.] Genotypes. *Int. J. Curr. Microbiol. App. Sci* 6(12): 945-951.
12. Kumar, N., Tokas, J., Kumar, P. and Singal, H. R. (2018). Effect of salicylic acid on postharvest quality of tomato (*Solanum lycopersicum* L.) Fruit. *International Journal of Chemical Studies*, 6(1): 1744-1747.
13. Gill, P., Singh, D., Kumar, R.T., Kumar, P. and Gupta, R.K. (2018). Comparative Effect of Different Fertilizers on Various Growth Parameters of *Lycopersicum esculantum*. *Int. J. Curr. Microbiol. App. Sci* 7(1): 56-60.

Popular Articles

1. Kumar, P., Midathala, R. and Kumar, N. (2017). Chromium toxicity tolerance and health risk assessment in a living system. *Agrobios newsletter*, XVI (06): 22-24.
2. Kumar, P., Kumar, N. and Midathala, R. (2017). Glycine betaine: A miraculous stress-relieving compound. *Agrobios newsletter*, XVI (04): 17-20.
3. Kumar, P., Midathala, R. and Kumar, N. (2018). Nanoparticles in controlling abiotic stress in crop plants. *Agrobios newsletter*, XVI (10):18-21.
4. Kumar, N., Midathala, R. and Kumar, P. (2018). Insight into the Molecular and Physiological Mechanism of Salt Tolerance of Halophytes. *Agrobios newsletter*, XVI (08): 12-13.

Conferences/Seminars Attended

1. Praveen kumar. Certificate of appreciation. National Conference on Trends in Nanobiotechnology (NCTN 2016). Dept. MBB&B, CCSHAU, Hisar, Haryana, India. 29th - 30th November 2016.
2. Praveen kumar, Naresh kumar, Jayanti tokas and H. R. Singal. Oral presentation on the effect of glycine betaine and AMF on alleviation of chromium stress in Sorghum plants. International conference on Emerging areas of Environmental science and Engineering, Department of environmental science, GJU, Hisar, Haryana, India. 16th – 18th February 2017.
3. Praveen kumar. Certificate of participation. Workshop on Capacity building on IPR instruments. HRM CCSHAU Hisar, India. 6th May 2017.
4. Praveen kumar, Jayanti tokkas, H. R. Singal and Parveen gill. Oral presentation on Glycine Betaine and Arbuscular Mycorrhizal Fungi Relieves Chromium Toxicity in Sorghum Plants by Interfering with Antioxidative Enzymes and Metabolites. 3rd International conference on bioresource and stress management. SIAM, Jaipur, Rajasthan, India. 8th – 11th November 2017.
5. Praveen kumar, Jayanti tokas, H.R. Singal, Parveen gill and Manohar lal. Effect of *Arbuscular mycorrhizal fungi* (AMF) on chromium toxicity in *Sorghum bicolor* L. International Conference on Microbes for health and wealth. Department of microbiology, MDU, Rohtak, Haryana, India. 14th November 2017.
6. Praveen Kumar, Jayanti Tokas, and H.R. Singal. Oral presentation on chromium (VI) heavy metal toxicity induced morphophysiological and biochemical changes in *Sorghum bicolor* L. International Conference on Global research initiatives for sustainable agriculture and allied sciences. (GRISAAS-2017) Rajasthan College of Agriculture, Maharana Pratap University of agriculture and technology, Udaipur, Rajasthan, India. 2nd – 4th December 2017.

7. Praveen kumar. Climate Impacts on Crop Yield and Food Security. National Seminar on Climate Change and Food Security. Centre for biotechnology, MDU Rohtak, Haryana, India. 25th January 2018.
8. Praveen kumar and Puneet beniwal. Role of Nanotechnology in controlling Plant growth and development under various stresses. International Conference on Bio and Nano Technologies for Sustainable Agriculture, Food, Health, Energy and Industry (21-23 February 2018) at GJU Hisar.
9. Praveen kumar and Puneet beniwal. Impact of environmental pollutant heavy metal chromium on plants growth and development. International conference on sustainable agriculture, energy, environment and technology (ICSAEET-2018) at 24-25th February 2018, MDU, Rohtak, Haryana.
10. Praveen kumar, Jayanti tokas and H R Singal. Oral presentation on Climate change causes a major loss of agriculture and food security in the world. National Seminar on “Emerging Trends and Challenges in Biosciences” at Gandhi memorial national college Ambala cant, Haryana. March 10, 2018.
11. Praveen kumar, Jayanti tokas and H.R. Singal. Importance of Nanomaterials (NMs) in Plant nutrition and its negative effects on the Ecosystem. International Conference on Microbial, Plant & Animal Genome (IC-MPAG 2018) at 29-31st March 2018, MODY University of science and technology, Sikar, Rajasthan.
12. Praveen kumar, Jayanti tokas, Naresh kumar and H. R. Singal. Oral presentation on the nutritive value of Sorghum crop for Animals and the effect of Heavy metal Chromium VI toxicity on it. Third National Conference on Contemporary Food Processing and Preservation Technologies (12-13 April 2018) at Shoolini University, Solan.
13. Praveen kumar, Jayanti tokas and H. R. Singal. Study of the antioxidative defence system of sorghum (*Sorghum bicolor* L.) under chromium VI toxic stress and its amelioration. International symposium on intrapreneurship: A need for sustainable agriculture (ISINSA-2019), February 2-3, 2019 at CCS Haryana Agricultural University, Hisar, (Haryana) India.
14. Praveen kumar, Jayanti tokas and H. R. Singal. Pre-sowing glycine betaine application as elicitor improves yield and quality of sorghum (*Sorghum bicolor* L.) under Cr VI toxic stress. International conference on climate change towards health and agricultural sustainability (CCHAS-2019), February 18-20, 2019 at the department of environmental science & engineering, GJU of science and technology, Hisar, Haryana, India.

Research Experience:

- Five year experience at CCSHAU of Next-generation sequencing, library preparation, sequencing, and data analysis. Plant molecular biology and peptide synthesis. Finding resources to help and prepare chemical synthesis.

- Experience of genomics, proteomics and metabolomics at CCSHAU academic level.
- Working experience of Biochemical techniques such as Characterization of Plasma Proteins by various physicochemical methods, Handling GC, HPLC, SDS-PAGE, ELISA, Kjeldahl /different protein estimation techniques and handling of lab equipment's such as Stability sample analysis by different methods and its data management, Analytical Method Qualification and Analytical Technology Transfer, Responsible for preparation of documentation of related activities.
- Worked on research problem entitled **“Biochemical studies on chromium toxicity and its amelioration in Sorghum (*Sorghum bicolor* L.)”** during PhD at CCS Haryana Agricultural University, Hisar (Haryana) India.
- Worked on research problem entitled **“A field study on Gene and Genotype frequency of autosomal and sex-linked Mendelian traits in humans”** for the fulfilment of the requirements for the degree of B.Sc (Biochemistry, Biotechnology, and Genetics) from Bangalore University, Bangalore (Karnataka) India
- Reviewer of the Journal Environmental chemistry letters and reviewed three articles.
- Reviewer of the Journal Biologia Plantarum.
- Reviewer of the International Journal of Phytoremediation.
- Reviewer of the Journal of Agricultural Science and Technology A & Journal of Agricultural Science and Technology B (USA).

Training/Workshops attended

1. Praveen kumar. Certificate of 21 days training on Tools and techniques in molecular biology bioinformatics and peptide synthesis. Dept. of animal biotechnology, LUVAS, Hisar, India. 4th – 25th September 2017.
2. Praveen kumar. Certificate of participation. Training programme in Statistical data analysis for the research scholar. Directorate of student welfare CCSHAU Hisar, India. 20th – 29th March 2017.
3. Praveen kumar. Certificate of participation. Workshop on Scientific / Technical writing. CCGHAU, Hisar, India. 18th – 19th April 2017.
4. Praveen kumar. Certificate of participation. Workshop on 50 years journey of Haryana: Media perspective. College of home science, CCSHAU, Hisar, India. 16th November 2017.

5. Praveen kumar. Certificate of participation. Workshop on Awareness and use of the Indian citation index (ICI) database. Nehru Library, CCSHAU, Hisar, India. 7th December 2017.
6. Praveen kumar. Online One week certificate course on Apiculture organised by department of Zoology, Balwant College, vita, Sangli, (M.S.) India from 13th July to 19th July 2020.

Key Skills:

- In-vitro assay development, validation and optimization of cell based assay and enzyme based assay for various targets in the respective therapeutic areas such as CNS/ Metabolic disorder/ Autoimmune/Inflammatory diseases/ Pain/Oncology in Drug Discovery field.
- Working extensively in the area functional cell based assays, validated the same to determine the activity of new chemical entities (NCEs).
- Responsible for writing of Protocols, uploading of the data, Lab note book record keeping, SOP writing.
- Analysis of experimental data using Graph pad PRISM (statistical Analysis Software) for IC50 and graphs.
- Designing experiments, execution, calculations, data analysis, graphical representations, report results to project teams.
- Resolving the complaints by finding the root cause analysis and taking the corrective actions.
- Follow the guidelines followed in the lab towards different safety aspects and help the lab head in designing the new guidelines.
- Follow ethical guidelines towards assay validation, data analysis and maintenance.
- Working in a team environment in order to deliver at a faster pace.

Language Proficiency:

- English (Read, Write and Speak)
- Hindi (Speak, Write and Speak)

Strengths

- ✓ Dedication, Hard Working and a Positive attitude.
- ✓ The faith, my parents have in me.
- ✓ Simplicity, honesty and patience.
- ✓ Believe in science and research.

Declaration

I hereby declare that the information provided by me is true to the best of my knowledge and belief.

Date: 16/02/2024

Place: Dahina(Haryana)



(Praveen Kumar)